Valerii Kliuchnikov

Senior Software Engineer

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SUMMARY

Senior Software Engineer (Go) with 9 years shipping production backend systems across fintech, AI tooling, and social platforms. Currently at Playneta building real-time LLM infrastructure at 9.6M messages/mo; previously delivered enterprise RBAC and payment systems at 2M+ daily transactions and 10K+ checks/sec with zero-downtime migrations for 1M+ users. Author of Engi, an open-source Go HTTP framework. Hands-on technical leadership: system design, 30+ technical interviews, 3 engineers mentored. Based in Barcelona, open to remote roles in EU and North America.

TECHNICAL SKILLS

Languages: Go, SQL, Python, TypeScript
Backend: Microservices, REST, gRPC, Protobuf, Gin, Echo, GORM, Wire DI, DDD, Domain-Driven Design, FastAPI, Strawberry
Data: PostgreSQL, ScyllaDB, Cassandra, Redis
Messaging: Kafka, Redpanda, RabbitMQ, asynq
AI/LLM: MCP, OpenRouter, multi-model routing, Agno, Claude Code, LLM inference pipelines
Cloud/DevOps: GCP (GKE, Pub/Sub), Kubernetes, Helm, Docker, GitLab CI, GitHub Actions
Observability: Prometheus, Grafana, OpenTelemetry, distributed tracing
Security: OAuth 2.0, OIDC, JWT, RBAC, authentication, authorization
Compliance: GDPR, PCI DSS
Practices: System design, technical leadership, code review, code refactoring, API development, mentoring, hiring

PROFESSIONAL EXPERIENCE

Playneta Limited — Nicosia, Cyprus (Remote)
Senior Software Engineer, B2B Contract — May 2026 - Present

- Owned system design and API development for a real-time LLM chat translation service in Go: approximately 9.6M messages per month at 612ms median and 981ms p95 end-to-end latency; auto-detection across 26 source languages; two-stage pipeline (Kafka consumer to asynq worker) hardened against hallucinations, truncation and concurrent updates.
- Cost-optimised two-model LLM routing via OpenRouter (Gemini 2.5 Flash Lite primary, GPT-4.1 Nano fallback) with per-model circuit breaker and Redis cross-message cache at 25% hit rate, keeping total LLM spend at approximately $591 per month (approximately $7K per year, 0.3% of platform revenue); full per-translation provenance in Cassandra and Kafka events.
- Translation infrastructure reliability: stage-2 LLM worker processes 5.1M tasks per week at 99.995% success rate; HTTP API serves 34M requests over 24 days at 99.998% success rate with p99 read latency of 183ms; Helm-based Kubernetes deployment with horizontal pod autoscaling across API, shared-consumer and worker pools.
- Built a platform-wide user-generated content moderation service from scratch using strict Domain-Driven Design layering (PostgreSQL, ScyllaDB, Redis, Wire dependency injection); rule-based engine (Aho-Corasick dictionary and regex for contacts and links) covering nicknames, bio, club metadata and chat messages across 5 content types; bulk dictionary API with versioning, per-field rejection codes and a userId-scoped check endpoint.
- Moderation service serves approximately 99,000 checks per day at p99 latency of 10ms, 10 times under the client fail-open deadline, with zero errors in 24 days of production; integrated into user, chat and clubs services so all text UGC and anti-spam routes through a single moderation path.

- Designed and shipped a reusable dead-letter queue layer for Redpanda consumers in the shared go-kiss2 library; adopted across 4 services (chat, clubs, translation, migration), eliminating silent message loss and capturing approximately 1,800 failed messages per day at a 0.016% failure rate into a single versioned DLQ topic for triage and replay.
- At-least-once delivery semantics with per-group self-replay, configurable ReplayDelay and MaxReplays, and a dedicated dlq.parked topic for exhausted messages; opt-in Permanent error marker lets handlers short-circuit non-retryable failures (schema violations, poison payloads) straight to parked, bypassing retry loops.

Refact.ai London, UK (Remote)
Software Engineer, B2B Contract August 2025 - April 2026

- Built MCP server infrastructure on Kubernetes (isolated pod-per-integration model), declarative JSON preset system (0 to 30+ connectors: Slack, GitHub, Notion, Stripe, Supabase, Discord, etc.), and in-chat setup flow — reducing integration onboarding from manual k8s configuration to minutes.
- Designed and shipped a complete OAuth 2.0 and token-based authorization framework covering 10+ external service providers (GitHub, Shopify, Google, and others).
- Authored flexus-client-kit — a Python SDK for agent developers with 15+ integration connectors (Gmail, Google Calendar/Sheets, Slack, GitHub, Jira, Telegram, Shopify, etc.) and abstractions for MCP tool registration, service lifecycle, and external auth flows.

Ozon Bank Moscow, Russia (Remote)
Senior Software Engineer May 2024 - July 2025

- Led system design and delivery of a Go-based RBAC platform (PostgreSQL, Redis, Kafka) serving 1M+ enterprise users under financial-regulator compliance; zero-downtime migration completed in 4 months, lifting permission-assignment success rate by 50%.
- Built an automated data-sync pipeline in Go for 200,000 legal entities using event-driven architecture (Kafka and gRPC), cutting manual data updates by 60% and reducing staleness by 75%.
- Architected a high-throughput notification service (Go, Kafka, PostgreSQL) processing 1M+ alerts per day at 99.9% uptime across multiple consumer groups.
- Extended RBAC with real-time transaction-limit enforcement at 10,000+ checks per second over 1M active accounts, reducing limit-breach errors by 90%; designed for horizontal scaling on Kubernetes.
- Conducted 30+ technical interviews, defined hiring rubrics, and mentored 3 engineers through structured onboarding, cutting ramp-up time by 40%.

Quadcode Larnaca, Cyprus (Hybrid)
Software Engineer May 2022 - May 2024

- Owned Go payment services layer (Redis, PostgreSQL, Kafka) processing up to 1M transactions per day across multiple crypto and fiat providers; optimised concurrency patterns cutting server load by 40% and reducing pipeline maintenance costs by 70%.
- Prototyped an advanced crypto payment provider integration, securing executive approval and rolling it out to three teams.
- Designed a declarative integration framework that reduced new payment-provider onboarding from 2 weeks to 1 week.
- Rebuilt GitLab CI with precommit hooks, accelerating pipelines by 20% and tightening code review cycles by 90%.
- Delivered GDPR-compliant anonymization and PCI DSS tokenization services from scratch.

ProCSy Moscow, Russia (Remote)
Software Engineer March 2020 - May 2022

- Built 10+ Hyperledger Fabric smart contracts, automating workflows and reducing manual reconciliation by 70%.
- Established an SMTP mailing system handling 10,000+ daily emails at 98% delivery rate.
- Decomposed a monolithic application into 6 microservices, cutting deployment time from 30 to 5 minutes.
- Automated CI pipelines with GitLab CI and golangci-lint, achieving 90% reduction in feedback cycles.

ITMO University Saint Petersburg, Russia (Hybrid)
Junior Software Engineer January 2018 - March 2020

- Developed a modular resource-visualization system in Go using object-oriented programming principles, and a GORM-based relational database schema for an internal simulation platform used in research courseware.

PROJECTS

Engi: Lightweight Go HTTP framework: declarative routing, structured services, automatic parameter validation, auth and CORS middleware, OpenTelemetry tracing, auto-generated OpenAPI documentation. Open-source; GoDoc, CI, Go Report Card. github.com/kliuchnikovv/engi

Nomi: End-to-end personal AI agent system: Go/Gin API gateway, Python/Agno agent orchestration (tool use, multi-agent teams), React Native mobile client. Plugin architecture — Gmail, Calendar, GitHub, Slack connectors. Deployed on Railway with Docker; 1,700+ commits. github.com/kliuchnikovv/nomi

Keystone: Local-first smart-home hub on ESP32 (C++/ESP-IDF): multi-chip architecture — brain (S3) runs entity registry, state machine and rules engine; radio co-processors handle Zigbee (C6), Matter and BLE. Go relay server, React Native mobile app over authenticated WebSocket/JSON protocol, OTA firmware. github.com/kliuchnikovv/keystone-esp

DnD: Detective RPG with a deterministic Go rules engine and an LLM narrator (OpenRouter/Anthropic). Trust-boundary architecture: model only narrates what the core decided. HTTP/WebSocket server, React Native mobile client (iOS-first), pluggable rule systems (D&D 5e, Cyberpunk RED). 188K lines Go, 1,211 tests, 10 ADRs. github.com/kliuchnikovv/dnd

EDUCATION

Bachelor of Science in Software Engineering September 2015 - June 2019
ITMO University, Saint Petersburg, Russia
Faculty of Software Engineering and Computer Systems

ADDITIONAL INFORMATION

Languages: Russian (Native), English (Professional working proficiency)
Interests: 3D modelling & printing, RC aircraft & cars, tabletop RPG, smart home, gaming